Assignment No: - 9

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Problem Statement

- Use appropriate algorithm and perform sentiment analysis for indian language.
- Objectives
- To understand the concept of sentiment analysis in Natural Language Processing (NLP).
- To study the challenges involved in processing Indian regional languages such as Hindi and Marathi.
- To learn and apply transformer-based algorithms (BERT / Multilingual BERT) for sentiment classification.
- To understand the role of tokenization and text preprocessing in improving model performance.
- To analyze how machine learning models classify text into positive, negative, and neutral sentiments.
- To evaluate the effectiveness of sentiment analysis in real-world applications like social media and reviews.

S/W Packages and H/W apparatus used

- **Operating System:** Windows / Linux / macOS
- **Programming Language:** Python 3.x
- **Development Tools:** Jupyter Notebook / Google Colab / VS Code
- **Library Used:** NLTK (Natural Language Toolkit)

Hardware Apparatus Used

- Computer or Laptop
- Processor: Intel / AMD or equivalent
- Minimum **8 GB RAM**
- Internet connection (for downloading NLTK resources)

THEORY :

Sentiment Analysis

Sentiment Analysis is a Natural Language Processing (NLP) technique used to determine the emotional tone or opinion expressed in a piece of text. It classifies text into categories such as Positive, Negative, or Neutral.

It is widely used in applications like:

- Social media analysis
- Customer feedback analysis
- Product reviews
- Chatbots and recommendation systems

Sentiment Analysis for Indian Languages

Indian languages such as Hindi, Marathi, Tamil, and Telugu have complex grammar, diverse vocabulary, and regional variations. Therefore, performing sentiment analysis on these languages requires models that can understand multilingual text.

Traditional methods like Bag of Words are less effective, so deep learning-based models like Transformers are used.

Algorithm Used: Transformer (BERT-based Model)

Transformer is a deep learning architecture that understands the context and meaning of words in a sentence. It is widely used in NLP tasks.

BERT (Bidirectional Encoder Representations from Transformers)

- BERT reads text in both directions (left to right and right to left)
- It understands the context of words better than traditional models
- It is pre-trained on large multilingual datasets

For Indian languages, Multilingual BERT (mBERT) or IndicBERT is used.

Working of the Algorithm

Step 1: Input Text

User provides text in any Indian language.

Example:
"मुझे यह बहुत अच्छा लगा"

Step 2: Tokenization

The input text is split into tokens using a tokenizer.

Example:
Tokens → [मुझे, यह, बहुत, अच्छा, लगा]

Step 3: Embedding

Each token is converted into numerical vectors so that the model can process it.

Step 4: Transformer Processing

The tokens are passed through multiple transformer layers where:

- Context is captured
- Relationships between words are learned

Step 5: Classification

The model predicts sentiment using a classifier layer.

Output:

- Positive
- Negative
- Neutral

Example Diagram

Input Text
"मुझे यह बहुत अच्छा लगा"
↓
Tokenization
↓
Transformer Model (BERT)
↓
Output: Positive Sentiment

Advantages of Using Transformer Models

- Handles multiple languages

- Understands context better
- Works well for complex sentences
- High accuracy compared to traditional methods

Conclusion

Sentiment analysis is an important NLP task used to understand user opinions. For Indian languages, transformer-based models like BERT provide accurate results by capturing contextual meaning. Using libraries such as Hugging Face Transformers makes implementation easy and efficient. This approach improves the performance of real-world applications like chatbots, review systems, and social media analysis.